

Convolutions

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1 An Convoluted Introduction

1.1 Image Processing

1.1.1 Defining Convolution for Images

Let us imagine you are a computer science student studying graphics. An important part of graphics is understanding matrices (if you are not failing the course, you are likely to know about these); they are what give structure to the pixels that make up an image.

Suppose that every pixel is an integer in the range $[0,255]$. We can then view an image as a large $n \times n$ matrix of pixels. Image processing has to do with changing the values of these pixels to produce a certain “edit” of this image. Blurring, for example, is one of the most widely used aspects of such pixel editing.

Before we get into convolutions, let us try to develop such an editing tool ourselves. If we focus on blurring specifically, we can quickly conclude that we cannot just focus one element of a list at a time. After all, how are we to blur by simply changing the RGB value of one pixel based on itself? If we were to transform a matrix in this way, it would simply scale the colors up and down, which, as a hypothetical graphics student, you hopefully see is not the process we are aiming for.

If we’re not going to be changing the RGB value by just some constant, then we must take some combination of neighbor pixels. This is (for the most part) how image processing works. Let us define a process by which this can be done.

Definition 1.1. Given a matrix called the **image**, a **kernel/convolution matrix** composes with the image to alter it.

For our purposes, the convolution matrix will be a 3×3 matrix.

This definition is rather vague, but the composition we are talking about here is the convolution operation, specifically for image processing, which we will define shortly.

As we mentioned earlier, if we are going to be editing one pixel at a time, we want to consider a neighborhood of points around the point we are trying to edit. Then, for a sufficiently large matrix of pixels, we can consider a small 3×3 submatrix as our image:

$$\begin{bmatrix} \ddots & \vdots & \vdots & \vdots & \ddots \\ \cdots & a_{11} & a_{12} & a_{13} & \cdots \\ \cdots & a_{21} & a_{22} & a_{23} & \cdots \\ \cdots & a_{31} & a_{32} & a_{33} & \cdots \\ \ddots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} = f$$

and apply a 3×3 convolution matrix on it. If our convolution matrix is

$$\begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \omega$$

then

$$f * \omega = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} b_{ij},$$

where $*$ is our convolution operation. It should be noted that we are not limiting the size of our image f , but we do limit the size of our convolution matrix ω .

1.1.2 Simple Example

Suppose that our image submatrix is

$$\begin{bmatrix} 46 & 30 & 13 \\ 0 & 23 & 34 \\ 5 & 1 & 4 \end{bmatrix} = f$$

and our convolution matrix is

$$\begin{bmatrix} 1 & 1 & 1 \\ -1 & 2 & 1 \\ -1 & -1 & -1 \end{bmatrix} = \omega$$

Clearly, we can only take the convolution matrix of a single point: in this case, f_{22} . Otherwise, we would need to use some sort of averaging or wrap-around technique to get additional points; for any f_{ij} , the only pair (i, j) in f that has 8 neighbors surrounding it (also called a **Moore neighborhood**) is f_{22} .

$$\begin{aligned} f[2, 2] * \omega &= (46)(1) + (30)(1) + (13)(1) + (0)(-1) + (23)(2) + (34)(1) + (5)(-1) + (1)(-1) + (4)(-1) \\ &= 159 \end{aligned}$$

So, without altering any of the other points, our new matrix f' would be $\begin{bmatrix} 46 & 30 & 13 \\ 0 & 159 & 34 \\ 5 & 1 & 4 \end{bmatrix}$.

1.1.3 More General Image Convolutions

Before we consider some properties of this convolution we defined, let's first slightly change our definition of convolutions to be more general. Previously, we had

$$f * \omega = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} b_{ij},$$

but we could also have rewritten this as

$$f[a, b] * \omega = \sum_{i=0}^3 \sum_{j=0}^3 f_{a+i-1, b+j-1} \omega_{i,j}$$

since we were only really talking about the center point anyway.

We'll use commas in the indices for ease of notation.

With this, we could then rewrite our standard definition of 3×3 convolutions between two matrices so that it returns a new matrix.

Definition 1.2. Let f be an $n \times n$ image matrix and ω be a 3×3 kernel. Let us define a 3×3 image convolution in the following way:

$$f \ast^* \omega = \delta,$$

for all indices a, b in f , where

$$\delta_{a,b} = f[a, b] \ast \omega.$$

We use $\ast^*$ here to explicitly separate this from our earlier definition of a convolution, which returns a single value, whereas $\ast$ returns an $n \times n$ array of elements.

Now that we've established a slightly updated definition, we can begin considering certain properties of this image convolution. The easiest to start off with is the *identity* kernel.

We know that the convolution works by considering a cell and its 8 Moore neighbors and mapping each of those to their corresponding places through a transformation due to the kernel. So, for some image f and kernel ω , the identity should go as follows:

$$f \ast^* \omega = f \implies f[a, b] \ast \omega = f_{a,b}, \forall a, b$$

and this is clearly true when we have a kernel

$$\omega_{\text{id}} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Now, let's consider something that is not necessarily a property of 3×3 convolutions, but is certainly important to this field: the idea of blurring. Since we are processing images, the blurring kernel should definitely come to mind. Since *blurring* is really just taking the average of all surrounding cells (as was mentioned earlier), we can deduce what a standard blurring kernel could look like:

$$\omega_{\text{blur}} = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Problem 1.1. Given an image f , its blurring kernel ω , and $f \ast^* \omega = \delta$, find a matrix κ such that $\delta \ast^* \kappa = f$.

The general form for a matrix convolution, if we were to extend both the kernel and image to some $m \times n$ size, is given by

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \ast \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \dots & b_{mn} \end{bmatrix} = \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} a_{(m-i)(n-j)} b_{(m+i)(n+j)}$$

1.2 Weighted Die and Lists

At first, what we do may seem a bit unmotivated. Eventually (hopefully), its use will become apparent.

Let us consider 12 total weighted dice. Assign each side some arbitrary probability, and form two horizontal lists with the 6 dice. A list below, going from side 1 to side 6 left to right, has been given some arbitrary probabilities for each side.

$$\begin{array}{cccccc} .22 & .5 & .08 & .04 & .12 & .04 \end{array} \quad (1)$$

$$\begin{array}{cccccc} .13 & .25 & .16 & .3 & .1 & .06 \end{array} \quad (2)$$

Let's consider how we would calculate the probability of rolling a certain sum with two dice. If we were to take our second list, reverse it, and shift it to the left, with .06 appearing under .22, then multiplying those probabilities together, we would find the total probability of rolling a 1.

$$\begin{array}{ccccccc} & & & .22 & .5 & .08 & .04 & .12 & .04 \\ .06 & .1 & .3 & .16 & .25 & .13 & & & \end{array}$$

This is what the list would look like if we were calculating the probability of rolling a sum of 4. Simply by looking at the list, we see that

$$P(4) = .22(.16) + .5(.25) + .08(.13).$$

To be more general, we can assign a variable to each of the probabilities.

$$\begin{array}{cccccc} a_1 & a_2 & a_3 & a_4 & a_5 & a_6 \end{array} \quad (3)$$

$$\begin{array}{cccccc} b_1 & b_2 & b_3 & b_4 & b_5 & b_6 \end{array} \quad (4)$$

with the obvious condition that the probabilities of each row sum to 1. So, then, for some $P(k)$ where $1 \leq k \leq 12$, we have that

$$P(k) = \sum_{i+j=k} a_i b_j$$

where we must make sure that a_i and b_j are in our rows.

2 Continuous Convolutions

2.1 An Introduction

Let us extend this notion to functions in general. Since we do not want to restrict ourselves to $\mathbb{Z}$, since most useful functions will be of $\mathbb{R}$, we'd like to take an infinite sum while still applying the "reverse list" idea.

Definition 2.1. The **convolution** $*$ of two functions f, g is

$$(f * g)(x) := \int_{-\infty}^{\infty} f(t)g(x-t) dt.$$

Remark: The German word for convolution is *faltung*, which becomes even more interesting later on.

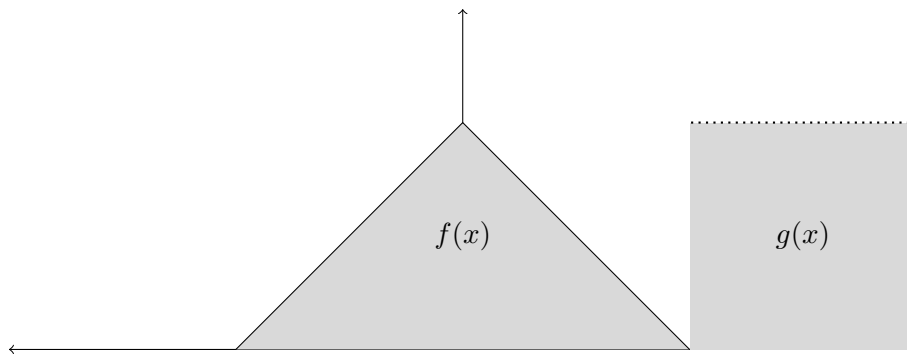
Before we get on to some basic properties of the convolution, let us consider what it actually does through practice.

For simplicity, let's define some simple piecewise functions. Let

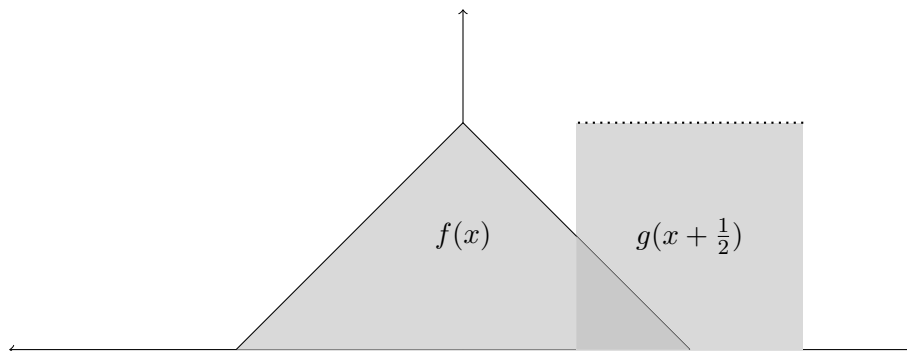
$$f(x) = \begin{cases} x + 1 & -1 \leq x \leq 0 \\ 1 - x & 0 \leq x \leq 1 \\ 0 & \text{otherwise,} \end{cases}$$

basically a simple triangle, and

$$g(x) = \begin{cases} 1 & 1 \leq x \leq 2 \\ 0 & \text{otherwise.} \end{cases}$$



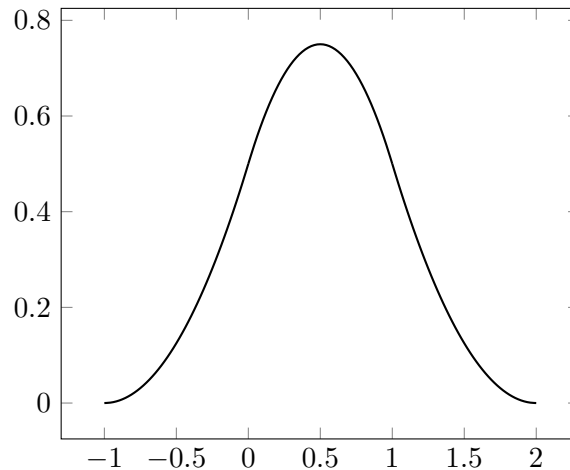
We know that we're going to get a function in terms of x . In this case, x is simply a translation of $g(x)$. For example, with $g(x + \frac{1}{2})$, we get the following:



So, as we change x , we translate $g(x)$ and find the new area of overlap between $g(x)$ and $f(x)$ —which happens to be the value of the convolution at x .

Therefore, we can calculate the resulting convolution function in a piecewise manner, as long as we consider *good* bounds for each function.

$$\begin{aligned}
&= \begin{cases} \int_{-1}^x t + 1 \, dt & -1 < x < 0 \\ \int_{-x}^0 t + 1 \, dt + \int_0^{1-x} 1 - t \, dt & 0 < x < 1 \\ \int_{x-1}^1 1 - t \, dt & 1 < x < 2 \end{cases} \\
&= \begin{cases} \frac{1}{2}(x+1)^2 & -1 < x < 0 \\ -\frac{x^2}{2} - \frac{1}{2}(1-x)^2 + 1 & 0 < x < 1 \\ \frac{1}{2}(x-2)^2 & 1 < x < 2 \end{cases}
\end{aligned}$$



This looks like quite the interesting graph. Often, it is useful to find the total area of the convolution; more specifically, defining

$$F := \int_{-\infty}^{\infty} (f * g)(x) \, dx.$$

In this case, we have to integrate by the small parts that we got. So,

$$F = \int_{-1}^0 \frac{1}{2}(x+1)^2 \, dx + \int_0^1 -\frac{x^2}{2} - \frac{1}{2}(1-x)^2 + 1 \, dx + \int_1^2 \frac{1}{2}(x-2)^2 \, dx,$$

giving us $F = \frac{1}{6} + \frac{2}{3} + \frac{1}{6} = 1$. This should, intuitively, make sense; after all, our rectangle g was sufficiently large to cover the entirety of f , which has an area of 1. However, it is true that 1 is merely a coincidence here. We will see through other examples that it is generally not going to be the area of the enclosed shape.

Problem 2.1. As one more introductory example, let's consider the convolution between two *box* functions. Similar to our previous $g(x)$, let's define

$$h(x) = \begin{cases} 1 & |x| \leq 1 \\ 0 & |x| > 1 \end{cases}$$

Evaluate $h * h$ and $\int_{-\infty}^{\infty} h * h$.

COMMUTATIVE PROPERTY

$$f * g = g * h.$$

Proof. This is rather trivial by substitution. In particular, substituting $u = x - t$, we get:

$$\begin{aligned} \int_{-\infty}^{\infty} f(t)g(x-t) dt &\xrightarrow{u=x-t} - \int_{\infty}^{-\infty} f(x-u)g(u) du \\ &= \int_{-\infty}^{\infty} f(x-u)g(u) du \end{aligned}$$

Because u has bounds of $-\infty$ and ∞ , the two expressions are equivalent in terms of t . $\square$

This should, intuitively, make sense. Superimposing one shape onto another and finding its overlap across all real numbers should be independent of which shape is superimposed onto which.

ASSOCIATIVE PROPERTY

$$(f * g) * h = f * (g * h).$$

Proof. By definition,

$$\begin{aligned} ((f * g) * h)(x) &= \int_{-\infty}^{\infty} (f * g)(t)h(x-t) dt \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u)g(t-u)h(x-t) du dt. \end{aligned}$$

By Fubini's theorem:

$$((f * g) * h)(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(u)g(t-u)h(x-t) dt du.$$

Let's consider the section $g(t-u)h(x-t)$; more specifically, $\int_{-\infty}^{\infty} g(t-u)h(x-t) dt$.

$$\begin{aligned} \int_{-\infty}^{\infty} g(t-u)h(x-t) dt &= \int_{-\infty}^{\infty} g((t+u)-u)h(x-(t+u)) dt \\ &= \int_{-\infty}^{\infty} g(t)h((x-u)-t) dt \\ &= (g * h)(x-u). \end{aligned}$$

Therefore,

$$\begin{aligned} ((f * g) * h)(x) &= \int_{-\infty}^{\infty} f(u)(g * h)(x-u) du \\ &= (f * (g * h))(x). \quad \square \end{aligned}$$

DISTRIBUTIVE PROPERTY

$$f * (g + h) = f * g + f * h.$$

Proof. By definition,

$$\begin{aligned}(f * (g + h))(x) &= \int_{-\infty}^{\infty} f(t)(g(x - t) + h(x - t)) dx \\ &= \int_{-\infty}^{\infty} f(t)g(x - t) dx + \int_{-\infty}^{\infty} f(t)h(x - t) dx \\ &= f * g + f * h. \quad \square\end{aligned}$$

Problem 2.2. Compute the convolution $f * g$, where $f(x) = e^{-ax^2}$, $g(x) = e^{-bx^2}$.

Before moving on to more interesting properties of the Dirichlet convolution, it is important to define an identity element. Seeing as how many properties are already fulfilled, it would be nice to fulfill this additional property. We can define one ourselves, and we will notice that it proves to be quite useful.

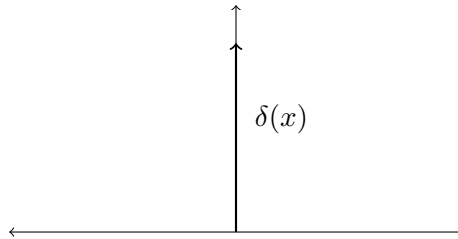
Definition 2.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$. The **Dirac delta function** δ is defined as

$$\delta(x) = \begin{cases} 0 & x \neq 0 \\ \text{undefined} & x = 0 \end{cases}$$

with the special notion that

$$\int_{-\infty}^{\infty} \delta(x) dx = 1.$$

Essentially, $\delta(x)$ looks as follows:



We should be able to see that

$$\int_{-\infty}^{\infty} f(t)\delta(x - t) dt = f(x)$$

just by intuition. We are essentially convoluting an infinitely tall line with a function f , so the overlap should return just that one section $f(x)$. The (slightly) more rigorous proof follows a similar trajectory; by considering $f(t)$ at such a short impulse simply as $f(x)$, we can take $f(x)$ out of the integrand.

$$\int_{-\infty}^{\infty} f(t)\delta(x - t) dt = f(x) \int_{-\infty}^{\infty} \delta(x - t) dt = f(x).$$

2.2 Solutions

Solution 2.1. An astute observer might notice that this is simply a box with a width of 2 and a height of 1. Setting up the appropriate integrals (spare me the image, please):

$$\begin{cases} \int_{-1}^{x+1} dt & 0 \leq x \leq 2 \\ \int_{x-1}^1 dt & -2 \leq x \leq 0 \end{cases} = \begin{cases} 2+x & 0 \leq x \leq 2 \\ 2-x & -2 \leq x \leq 0 \end{cases}$$

Then,

$$F = \int_{-\infty}^{\infty} h * h = \int_0^2 2+x \, dx + \int_{-2}^0 2-x \, dx$$

$$F = \boxed{12}.$$

Solution 2.2. Let us set up the computation.

$$\begin{aligned} f * g &= \int_{-\infty}^{\infty} e^{-at^2} \cdot e^{-b(x-t)^2} \, dt \\ &= e^{-bx^2} \int_{-\infty}^{\infty} e^{-(a+b)t^2 + 2bxt} \, dt \end{aligned}$$

Then, by completing the square,

$$\begin{aligned} f * g &= e^{-bx^2} e^{\frac{b^2 x^2}{a+b}} \int_{-\infty}^{\infty} e^{-\left(t\sqrt{a+b} - \frac{bx}{\sqrt{a+b}}\right)^2} \, dt \\ &\xrightarrow{u=t\sqrt{a+b}} \frac{e^{\frac{b^2 x^2 - b^2 x^2 - abx^2}{a+b}}}{\sqrt{a+b}} \int_{-\infty}^{\infty} e^{-u^2} \, du \\ &= \boxed{e^{-\frac{abx^2}{a+b}} \sqrt{\frac{\pi}{a+b}}} \end{aligned}$$

3 Dirichlet Convolutions

3.1 Why *Limit* Ourselves?

I am a huge fan of number theory, much more than I am a fan of calculus. As such, this section will have more problems created by me or problems that are widely available and considered and cannot be attributed to a single author or source.

Looking at convolutions across arithmetic functions may seem like a step down, but I assure you that it's also an interesting topic.

3.2 Arithmetic Functions

Definition 3.1. An **arithmetic function** is a function $f : \mathbb{N} \rightarrow \mathbb{C}$.

Remark: For our purposes, let $\mathbb{N}$ include 0. Not all arithmetic functions are well defined at 0, so most will specify their value at 0.

Often, an arithmetic function is defined to exploit certain properties of natural numbers. It is most common for these functions to be strictly $\mathbb{N} \rightarrow \mathbb{N}$. An example of such an arithmetic function that most are aware of is τ , where

$$\tau(n) = \text{number of positive divisors of } n$$

which also has a well-known explicit form.

To further qualify certain arithmetic functions, we must know some of their properties.

Definition 3.2. A function is **multiplicative** iff f is an arithmetic function and $f(a)f(b) = f(ab)$, $\forall a, b : \gcd(a, b) = 1$.

A function is **totally multiplicative** iff f is an arithmetic function and $f(a)f(b) = f(ab)$, $\forall a, b$.

These definitions imply that for a multiplicative function f , $f(1) = 1$. This is not necessarily true for the case where $f(n) = 0$ for some positive integer $n > 1$, but it holds otherwise. Though the converse is not necessarily true, it is helpful to check the base case of $n = 1$ when determining whether an arithmetic function is multiplicative or not. If $f(1) \neq 1$, it is clear that the function is not multiplicative.

The divisors function τ is a multiplicative function. We can prove this quite easily by taking two coprime positive integers n_1 and n_2 . Let $n_1 = p_1^{r_1} p_2^{r_2} \dots p_k^{r_k}$ and $n_2 = p_{k+1}^{r_{k+1}} \dots p_n^{r_n}$.

It's clear from here that

$$\tau(n_1) = (r_1 + 1)(r_2 + 1) \dots (r_k + 1)$$

$$\tau(n_2) = (r_{k+1} + 1)(r_{k+2} + 1) \dots (r_n + 1)$$

and thus $\tau(n_1)\tau(n_2) = \tau(n_1 n_2)$.

However, this is not enough to demonstrate that the function itself is multiplicative. Indeed, the explicit formula essentially tells us that the function is multiplicative (and is based on that claim), and we will generally not have the explicit formula for some function, so we will need to work with the little information that we have.

We can explicitly demonstrate the multiplicativity of τ . If we write τ as a summation function:

$$\tau(n) = \sum_{d|n} 1.$$

If we choose some $n_1, n_2 : \gcd(n_1, n_2) = 1$,

$$\tau(n_1) = \sum_{d|n_1} 1.$$

$$\tau(n_2) = \sum_{d|n_2} 1.$$

Given that $\gcd(n_1, n_2) = 1$, we can rewrite this as,

$$\tau(n_1)\tau(n_2) = \sum_{d|n_1} 1 \sum_{d|n_2} 1 = \sum_{d|n_1 n_2} 1 = \tau(n_1 n_2).$$

Most useful and interesting arithmetic functions are *not* totally multiplicative, and the most common totally multiplicative ones are trivial. $f \equiv 1$ is an intuitive one, though it is clear to see that $f(n) = n$ and $f(n) = n^k$ also satisfy the condition.

A non-trivial arithmetic function that retains the property of total multiplicativity is the **Louiseville function**. The Louiseville function $\lambda : \mathbb{N} \rightarrow \mathbb{N}$ is defined as

$$\lambda(n) = (-1)^{\Omega(n)}$$

where $\Omega(n) = \sum_{p^\alpha | n} 1$ and $\Omega(1) = 0$. It's pretty simple to prove multiplicativity for this function once considering that Ω is **additive** for all a, b : $\Omega(ab) = \Omega(a) + \Omega(b)$, which is intuitive but can be proven simply.

Then,

$$\lambda(a)\lambda(b) = (-1)^{\Omega(a)+\Omega(b)} = \lambda(ab)$$

.

Definition 3.3. The **divisor sum** of an arithmetic function f is

$$[D(f)](n) = \sum_{d|n} f(d)$$

It may alternatively be written as $\sigma_k = \sum_{d|n} f(d)^k$, but given that we are working with $k = 1$, it will likely be written the first way.

As a rudimentary example, consider $[D(f)](n)$ where $f(n) = n$. Then,

$$[D(f)](n) = \sum_{d|n} f(d) = \sum_{d|n} d$$

This is equivalent to the sum of divisors function σ , which has a relatively well-known explicit form.

Such is often true with divisor sums: divisor sums with well-known functions often produce other less trivial well-known functions.

Problem 3.1. Let λ be the Louiseville function. Compute $[D(\lambda)](n)$ for positive integer n .

Problem 3.2. Let λ be the Louiseville function. Compute

$$\sum_{n=1}^N \lambda(n) \left\lfloor \frac{N}{n} \right\rfloor$$

for positive integers n, N .

Problem 3.3. Define the **Möbius function** μ as

$$\mu(n) = \begin{cases} (-1)^k & \text{if } n = p_1 \dots p_k, k \text{ distinct primes} \\ 0 & \text{otherwise} \end{cases}$$

for positive integer n and let $\mu(1) = 1$. Find $[D(\mu)](n)$.

3.3 The Dirichlet Convolution

3.3.1 Definition and Properties

Definition 3.4. A **Dirichlet convolution** is an operation $*$ on two arithmetic functions $f, g : \mathbb{N} \rightarrow \mathbb{C}$ such that

$$(f * g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right)$$

Problem 3.4. Compute

$$\sum_{d|n} \varphi(d)$$

for positive integer n .

There are some interesting and fundamental properties to the Dirichlet convolution once we define some basic and useful functions. Define $e, \mathbb{1} : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$\mathbb{1}(n) \equiv 1$$

$$e(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases}$$

Assuming f, g are arithmetic functions, the following properties hold:

COMMUTATIVE PROPERTY

$$f * g = g * f$$

Proof. Commutativity is the easiest of the properties to prove. Consider that for each pair $\{d, \frac{n}{d}\}$, there will also be a pair $\{\frac{n}{d}, d\}$, so every term will appear in both $f * g$ and $g * f$.

ASSOCIATIVE PROPERTY

$$(f * g) * h = f * (g * h)$$

Proof.

FUNCTION IDENTITY

$$f * e = f = e * f$$

Proof.

$$(f * e)(n) = \sum_{d|n} f(d)e\left(\frac{n}{d}\right)$$

and to get $e(1)$, we must have $d = n$, so the convolution becomes $f(n)$.

SUMMATION IDENTITY

$$f * \mathbb{1} = Df = \mathbb{1} * f$$

Proof.

$$(f * \mathbb{1})(n) = \sum_{d|n} f(d) \mathbb{1}\left(\frac{n}{d}\right) = \sum_{d|n} f(d) = [D(f)](n).$$

INVERSE

$$\mu * \mathbb{1} = e$$

Proof. First, for $n = 1$:

$$(\mu * \mathbb{1})(n) = \mu(1) \mathbb{1}(1) = 1 = e(1).$$

Then, for $n > 1$:

$$(\mu * \mathbb{1})(n) = [D(\mu)](n) = 0 = e(1).$$

Theorem 3.1 (Möbius Inversion). For an arithmetic function f :

$$f = \mu * Df$$

where μ is the Möbius function.

Proof. Using the properties of the Dirichlet Convolution, the derivation is straightforward.

$$\mu * Df = \mu * \mathbb{1} * f = e * f = f$$

□

Problem 3.5. Compute $\varphi(n)$ for positive integer n using the result of the previous problem and Möbius inversion.

Problem 3.6. Compute

$$\sum_{d|n} \varphi(d) \sigma\left(\frac{n}{d}\right)$$

for positive integer n .

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3.4 Solutions

Solution 3.1. We know that the Louiseville function is completely multiplicative. Therefore, if we evaluate $[D(\lambda)](n)$ for some $n = p^k$, we can generalize it to all positive integer n . Making this

substitution,

$$[D(\lambda)](p^k) = \sum_{d|n} \lambda(p^k) = \lambda(p)^k + \lambda(p)^{k-1} + \dots + \lambda(1).$$

It's clear that $\lambda(p) = -1$, so $\lambda(p^j) = (-1)^j$. Therefore, $[D(\lambda)](p^k) = 1$ when $k \mid 2$ and 0 otherwise.

$$[D(\lambda)](n) = \begin{cases} 1 & \text{if } \exists k \in \mathbb{N} \mid k^2 = n \\ 0 & \text{otherwise} \end{cases}$$

Solution 3.2. Notice that the given equation is equivalent to

$$\sum_{n=1}^N \sum_{d|n} \lambda(d).$$

Using the previous problem, we have an easy closed form for $\sum_{d|n} \lambda(d) = [D(\lambda)](n)$. Then, we can rewrite this once again as

$$\sum_{\substack{d=1 \\ n^2=d}}^N 1$$

for $n \in \mathbb{N}$. This is equivalent to counting the number of squares less than N , or $\lfloor \sqrt{N} \rfloor$.

$$\sum_{n=1}^N \lambda(n) \left\lfloor \frac{N}{n} \right\rfloor = \lfloor \sqrt{N} \rfloor$$

Solution 3.3. First, consider the case where $n = 1$. Then,

$$[D(\mu)](1) = \mu(1) = 1$$

For $n > 1$, notice that only terms containing distinct primes are counted; otherwise, they evaluate to 0. Thus, we can group the sum by the number of distinct primes in each function:

$$\begin{aligned} [D(\mu)](n) &= 1 + (\mu(p_1) + \dots + \mu(p_k)) + (\mu(p_1 p_2) + \dots + \mu(p_{k-1} p_k)) + \dots + \mu(p_1 \dots p_k) \\ &= \sum_{i=0}^k (-1)^i \binom{k}{i} \\ &= (1 - 1)^k = 0 \end{aligned}$$

$$[D(\mu)](n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases} = e(n)$$

Solution 3.4. We claim that

$$\sum_{d|n} \varphi(d) = \sum_{d|n} \varphi\left(\frac{n}{d}\right),$$

and we will eventually see that this makes our reasoning easier. We can almost immediately see that this is true; for any $\varphi\left(\frac{n}{d}\right) = k$, $k \mid n$, so k is also a divisor. Since the LHS is summing over all divisors, they must be the same sum.

First, consider the sequence

$$\left\{\frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}, \frac{n}{n}\right\}.$$

There are clearly n elements in this set.

Consider the sequence of reduced fractions where $d \mid n$ and d is fixed

$$\left\{\frac{a_1}{d}, \frac{a_2}{d}, \dots, \frac{a_i}{d}\right\},$$

where $\gcd(a_i, d) = 1$, and there are $\varphi(d)$ elements in this set. Then, the result follows.

$$\boxed{\sum_{d \mid n} \varphi(d) = n}.$$

Solution 3.5. This problem is relatively straightforward using Möbius Inversion.

$$\varphi(n) = (\mu * D[\varphi])(n) = \boxed{\sum_{d \mid n} \mu(d) \left(\frac{n}{d}\right)}.$$

It is also possible to write this in other forms, but that is up to the wonderful reader to find.

Solution 3.6. Observe that this is a convolution, specifically

$$(\varphi * \sigma)(n).$$

Since $\sigma = \text{id} * \mathbb{1}$ and $\varphi * \mathbb{1} = \text{id}$,

$$\sigma * \varphi = (\text{id} * \mathbb{1}) * \varphi = \text{id} * (\mathbb{1} * \varphi) = \text{id} * \text{id}.$$

$$\implies \sum_{d \mid n} \varphi(n) \sigma\left(\frac{n}{d}\right) = (\text{id} * \text{id})(n) = \sum_{d \mid n} d \cdot \frac{n}{d} = n\tau(n).$$

$$\boxed{\sum_{d \mid n} \varphi(d) \sigma\left(\frac{n}{d}\right) = n\tau(n)}$$

where $\tau(n)$ is the number of divisors of n , which is easily computable given n , and thus this is a nice reduced form.

Alternative Solution. This alternative solution requires knowing the closed forms of both $\varphi(n)$ and $\sigma(n)$, which may not be the most accessible to those not accustomed to working with such functions. The preferred solution is the one above.

Recall that

$$\varphi(n) = n \prod_i \frac{p_i - 1}{p_i}$$

$$\sigma(n) = \prod_i \frac{p_i^{a_i+1} - 1}{p_i - 1}$$

Let $n = p^k$. We don't need a more complicated value of n because we know that both functions are multiplicative, and thus the convolution is also multiplicative. Then,

$$\begin{aligned}
\sum_{d|n} \varphi(d) \sigma\left(\frac{n}{d}\right) &= p^k(\sigma(p^k)\varphi(1) + \sigma(p^{k-1})\varphi(p) + \dots + \sigma(1)\varphi(p^k)) \\
&= \frac{p^{k+1} - 1}{p - 1} + p^k - 1 + (p^{k-1} - 1)(p) + (p^{k-2} - 1)(p^2) + \dots + p^k + p^{k-1} \\
&= 1 + p + p^2 + \dots + p^{k-1} - 1 - p - p^2 - \dots - p^{k-1} + (k+1)p^k \\
&= (k+1)n.
\end{aligned}$$

For more complicated values of n , such as $n = p^\alpha p^\beta \dots$, we will have $n(\alpha+1)(\beta+1)\dots$, which is equivalent to $\boxed{n\tau(n)}$.